Signals & Systems Solutions — Solutions (SIGNALS-SYSTEMS-HS2017)

Q#	Question / Context	Selected Answer	Solution Logic
q1	Exact discretization of the continuous-time system with sampling time Ts	**q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts (1 + Ts/2)], [Ts]] u[n]**	*Compute matrix exponential Φ = exp(A Ts) = I + A Ts because A^2 = 0, giving [[1, Ts],[0,1]]. Compute Γ = ∫0^Ts exp(A τ) dτ B = (Ts I + A Ts^2/2) B = [[Ts + Ts^2/2],[Ts]] = [[Ts (1 + Ts/2)],[Ts]].*
q2	Forward Euler discretization of the continuous-time system with sampling time Ts	**q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts], [Ts]] u[n]**	*Forward Euler yields q[n+1] = (I + Ts A) q[n] + Ts B u[n]. I + Ts A = [[1, Ts],[0,1]] and Ts B = [[Ts],[Ts]].*
q3	Number of periods of the sampled cosine u[n] = cos(pi/100 n) over N=1000 samples	**5**	*Sampling u(t)=cos(pi/10 t) with Ts=0.1 gives u[n]=cos(pi/100 n). The discrete-time angular frequency is ω0=pi/100 rad/sample, so the fundamental period is N0=2π/ω0=200 samples. Over N=1000 samples, the number of periods is 1000/200=5.*
q4	Fundamental period of the discrete-time signal x[n] when sampling the periodic continuous-time signal x(t) with Ts=0.5	**4**	*From the provided plot, x(t) repeats every T0=2 seconds (e.g., identical waveform from t=-2 to t=2 and again from t=2 to t=4). Sampling with Ts=0.5 yields N0 = T0/Ts = 2/0.5 = 4 samples per period, which is the fundamental period of {x[n]}.*
q5	Determine the discrete-time signal obtained by sampling the given periodic continuous-time signal with Ts=1, first sample at t=0.	**x[n] = 0.5 + cos(pi/4 n) - 0.3 sin(3 pi n)**	*The CT signal appears to be x(t)=0.5+cos(π/4 t) (DC 0.5, amplitude 1, period 8). Sampling at t=n yields x[n]=0.5+cos(π/4 n). The term -0.3 sin(3π n) equals zero for integer n, so option A matches the sampled signal.*
q6	Find which discrete-time sequence equals the sampled version of x(t)=cos(6π/5 t)+cos(2π t) with Ts=1 and t0=0.	**x_2[n]**	*Sampling gives x[n]=cos(6π/5 n)+cos(2π n). Since cos(2π n)=1 for integer n, x[n]=1+cos(6π/5 n), which exactly matches option x_2[n].*
q7	Sampling x(t) with Ts=1 and t0=0.25	**x3[n]**	*x[n] = cos(6π/5 (n+0.25)) + cos(2π (n+0.25)) = cos(6π/5 n + 3π/10) + cos(2π n + π/2) = cos(6π/5 n + 3π/10) + 0 = x3[n]*
q8	Select all linear systems among G1-G4	**G1, G2, G3**	*G1: y[n]=u[|2n|]x[n]=x[n] linear; G2: y[n]=(u[n]+u[n+1])x[n] linear time-varying gain; G3: linear difference equation; G4: nonlinear due to (y[n-1])^2 term.*
q9	Time-invariance of systems G1-G4	**G_2, G_3, G_4**	*G1 output is constant (independent of input) but not shift-invariant because a shifted input does not produce a shifted output. G2 can be expressed as y[n]=x[n]+x[n+1] with x[n]=u[n], which is time-invariant. G3 is an LTI difference equation with constant coefficients, hence time-invariant. G4's rule y[n]=0.5[y[n-1]]^2+u[n] contains no explicit n dependence, making it time-invariant despite nonlinearity.*
q10	BIBO stability of systems G1-G4	**G_1, G_2, G_3**	*G1 and G2 produce bounded deterministic outputs for any bounded input, thus BIBO stable. G3 is an LTI system with poles at z=0.5 (double), inside the unit circle, guaranteeing BIBO stability. G4 exhibits quadratic growth; with a constant unit-step input the output diverges, so it is not BIBO stable.*
q11	Stability of three given systems: G1 FIR impulse response, G2 first-order difference equation with pole at 2 (anti-causal), G3 second-order system with poles at magnitude 0.5.	**G_1, G_2, G_3**	*G1 is FIR => absolutely summable => stable. G2 has pole at z=2; anti-causal ROC |z|<2 includes unit circle => stable. G3 poles at 0.25±j0.433 magnitude 0.5 <1 => stable. Hence all three are stable.*
q12	Which step response(s) could come from an FIR filter, given plots of y1[n] and y2[n] for n=-1..7 to unit step input.	**{y_1[n]}**	*An FIR filter's step response becomes constant after the filter length; only y1[n] shows such settling behavior, while y2[n] continues to change, indicating an IIR response.*
q13	Select the difference equation that produces the step response y■[n] shown in the figure (n = -1..7).	**y[n] = 1/2 y[n-1] + u[n]**	*Assuming zero initial rest, compute the step response of each candidate. For y[n] = 0.5 y[n-1] + u[n], the solution is y[n] = 2 - 0.5■ for n ≥ 0, giving y[0]=1, y[1]=1.5, y[2]=1.75, approaching 2. This matches the shown monotonic increase from 1 at n=0 toward 2, while other options produce different initial values or behaviors (e.g., starting at 0, 1.5, 2 with oscillation or drop).*
q14	Select the difference equation that produces the step response y■[n] shown in the figure (n = -1..7).	**y[n] = 2 u[n] - 1/2 u[n-1]**	*For the FIR form y[n] = 2u[n] - 0.5u[n-1], the step response is y[0]=2, y[1]=1.5, and y[n]=1.5 for n≥2. This matches the figure's response that starts at 2 at n=0, drops to 1.5 at n=1, and remains constant thereafter, unlike the other options which either start at different values, oscillate, or grow without bound.*
q15	Which output sequence y corresponds to y = h_3 * u?		*The figure showing the four output sequences y_A, y_B, y_C, y_D is required to determine which matches the convolution of h_3 with u; without it, the answer cannot be selected.*
q16	Which impulse response h corresponds to y_B = h * u?		*The figure showing the four output sequences y_A, y_B, y_C, y_D is required to determine which impulse response (h_1..h_4) produces y_B when convolved with u; without it, the answer cannot be selected.*

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q17	Select all stable DT LTI systems from the given state-space descriptions.	**G_1, G_3, G_4, G_5**	*Stability requires all eigenvalues of A to have magnitude <1. G1: eigenvalues 0,0,0 -> stable. G2: eigenvalue 1 -> not stable. G3: eigenvalues ±1/√2 ≈0.707 -> stable. G4: eigenvalues 0.5 and 0 -> stable. G5: eigenvalue 0 -> stable. Hence stable systems are G1, G3, G4, G5.*
q18	Select all FIR systems from the given state-space descriptions.	**G_1, G_5**	*FIR iff A is nilpotent (some power equals zero). G1: strictly upper triangular -> A^3=0 -> FIR. G2: A=1 not nilpotent -> not FIR. G3: eigenvalues ±0.707 not zero -> not nilpotent. G4: eigenvalues 0.5 and 0 -> not nilpotent. G5: A=0 nilpotent -> FIR. Thus FIR systems are G1 and G5.*
q19	Find state-space representation of system with LCCDE y[n] = 1/2 y[n-2] + u[n] + u[n-1]	**G_3**	*Derive H(z) = (z^2+z)/(z^2-1/2). Compute each option's transfer function; only G3 yields H(z)=1+(z+1/2)/(z^2-1/2) which simplifies to the same.*
q20	Identify which system has transfer function H(z) = z + (1/2 z)/(z - 1/2)	**G_6**	*Simplify to H(z)=z^2/(z-1/2). The impulse response includes a non■causal δ[n+1] term; only G6's shown impulse response matches this pattern.*
q21	Steady-state value of y[n] as n→∞ for unit step input	**4**	*System is stable with poles at 0.5. DC gain H(1)=z^{-1}/(1–z^{–1}+¼z^{–2})|_{z=1}=1/(1–1+¼)=4. For unit step input, y[∞]=4·1=4.*
q22	Output for input u[n]=½ cos(π n)	**y[n] = 2/9 cos(pi n - pi)**	*Frequency response H(e^{jω})=e^{–jω}/(1–e^{–jω}+¼e^{–j2ω}). At ω=π, e^{–jπ}=–1, e^{–j2π}=1 → H=–1/(1+1+¼)=–4/9. Magnitude 4/9, phase π. Input amplitude ½ gives output amplitude (½)(4/9)=2/9 and phase shift π → y[n]=(2/9)cos(π n+π)=(2/9)cos(π n–π).*
q23	Output of cascade G_A = G_* -> G_* for input u[n] = 2 cos(π/2 n) + 3/2 cos(2π/3 n) + sin(π n)	**y[n] = -2/9 cos(pi/2 n) + 1/9 sin(pi n)**	*The cascade squares the magnitude and doubles the phase: |H_A| = |H|^2, ∠H_A = 2∠H. From the frequency response plots (implied), |H(π/2)| = 1/3, ∠H(π/2) = π/2, and |H(2π/3)| = 0. Thus the π/2 component becomes 2·(1/3)^2 cos(π/2 n + π) = -2/9 cos(π/2 n). The 2π/3 component vanishes. The sin(π n) term is zero for integer n.*
q24	Output of parallel sum G_B = G_* + G_* for the same input	**y[n] = 4/3 cos(pi/2 n - pi/2) + 2/3 sin(pi n)**	*The parallel sum doubles the magnitude and keeps the phase: |H_B| = 2|H|, ∠H_B = ∠H. Using |H(π/2)| = 1/3, ∠H(π/2) = π/2, and |H(2π/3)| = 0, the π/2 component becomes 2·2·(1/3) cos(π/2 n + π/2) = 4/3 sin(π/2 n) = 4/3 cos(π/2 n - π/2). The 2π/3 component vanishes. The sin(π n) term is zero.*
q25	Select magnitude plot matching causal sequence x[n] (stem plot n=0..10)		*Figure of magnitude plots and stem plot not provided; cannot determine which option matches.*
q26	Select magnitude plot matching another causal sequence x[n] (stem plot n=0..10)		*Figure of magnitude plots and stem plot not provided; cannot determine which option matches.*
q27	Identify transfer function of LTI system G from its magnitude and phase response.	**H(z) = (z + 1)^2 / (4 z^2)**	*The magnitude is unity at DC, drops to zero at Nyquist, and the phase is linear with slope –1 (–ω rad). This corresponds to a double zero at z = –1 and a double pole at the origin, giving H(z) = (z+1)^2/(4z^2).*
q28	Select magnitude/phase plot of system G cascaded with a unit delay.	**Plot B**	*Cascading with a delay multiplies the transfer function by z■¹, which leaves the magnitude unchanged and subtracts ω from the phase. Plot B shows the same magnitude as G but with a phase shifted downward by ω, matching this effect.*
q29	DFS magnitude of scaled periodic sequence	**|X[1]| = 6|k|**	*Using DFS definition X[k] = Σ x[n] e^{-j2πkn/N} (no 1/N factor) with N=6 and x[n]=k[-1,-2,-1,1,2,1], the sum for k=1 yields X[1]=k(-3+j5.196) whose magnitude is 6|k|.*
q30	DFT coefficient at bin 1 for sum of two complex exponentials	**X[1] = 0**	*The signal x[n]=2e^{j4π/7 n}+2e^{j10π/7 n} has period N0=7. With DFT length N=21=3·N0, nonzero bins occur at k=m·L where m=2,5 and L=3, i.e., k=6 and 15. Hence X[1]=0.*
q31	DFT coefficient X[6] for x[n]=2e^{j4π/7 n}+2e^{j10π/7 n} with N=21	**X[6] = 42**	*The DFT sum splits into two geometric series. For the first term, ω1 - 2π·6/21 = 0, giving sum =21 → contribution 2·21=42. For the second term, ω2 - 2π·6/21 = 6π/7 ≠ 0 mod 2π, and the series sums to zero because r^21=1. Hence X[6]=42.*
q32	Given |X[k]|={0,2,0,2} for N=4, which plot shows |X[k]| for N=6 of the same real signal?	**Plot C**	*The N=4 magnitude indicates a cosine x[n]=cos(π n/2). Computing the 6■point DFT of x[n] for n=0…5 yields |X[k]| = {1,2,2,1,2,2}. The plot matching this symmetric pattern (two lower bins at k=0,3 and four higher bins at k=1,2,4,5) is Plot C.*
q33	Impulse response of H(z) = (z^2 + 0.5z)/(z^2 - 0.25)	**h = {..., 0, 1 (n=0), 1/2, 1/4, 1/8, ...}**	*Simplify H(z) = z/(z-1/2). For causal ROC |z|>1/2, inverse Z-transform yields h[n] = δ[n] + (1/2)^n u[n-1]. Thus h[0]=1 and h[n]=(1/2)^n for n≥1, matching the given option.*

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q34	Magnitude response of H(z) = 1/8 (z-1)/(z^2+z+0.5)	**Plot C**	*Zero at z=1 gives null at DC (ω=0). Poles at (-0.5±0.5j) (radius ≈0.707, angle ±135°) create peaks near ω=±3π/4. Magnitude rises from zero at ω=0 to peaks then falls to 0.5 at ω=π, which corresponds to Plot C.*
q35	Causal LTI system pole-zero plot; identify possible magnitude response.	**Plot D**	*For a causal stable system all poles must lie inside the unit circle. The magnitude response that shows low attenuation at low frequencies and a roll-off consistent with the given pole-zero locations corresponds to Plot D.*
q36	Frequency transformation converting a CT low-pass filter to a high-pass filter.	**H_HP(s) = H_LP(10/s)**	*The standard low-pass to high-pass transformation is H_HP(s) = H_LP(Ω_c / s), where Ω_c is the desired cutoff frequency. From the Bode plots the cutoff appears at 10 rad/s, giving H_HP(s) = H_LP(10/s).*
q37	Select G1, G2, G3 classes to realize a band-stop filter using series-parallel sum.	**G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_HP(z)**	*Overall transfer function G(z)=H1(z)H2(z)+H3(z). With H1=LPF, H2=1, H3=HPF we get G(z)=LPF(z)+HPF(z). Choosing non-overlapping cutoff frequencies for the LPF and HPF makes the sum pass low and high frequencies while attenuating the middle band, yielding a band-stop response.*
q38	Select G1, G2, G3 classes to realize a band-pass filter using series-parallel sum.	**G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 0**	*Overall transfer function G(z)=H1(z)H2(z)+H3(z). With H1=LPF, H2=HPF, H3=0 we get G(z)=LPF(z)·HPF(z). The cascade passes only frequencies that are both below the LPF cutoff and above the HPF cutoff, i.e., a band between the two cutoffs, producing a band-pass characteristic.*
q39	Determine gain values for block diagram realization of given second-order IIR filter.	**g_0 = b_0, g_1 = b_1, g_2 = b_2, g_3 = 1, g_4 = -a_1, g_5 = -a_2**	*From the block diagram, y[n] = g3*(g0 u[n] + g1 u[n-1] + g2 u[n-2] + g4 y[n-1] + g5 y[n-2]). Matching coefficients to y[n] = -a1 y[n-1] - a2 y[n-2] + b0 u[n] + b1 u[n-1] + b2 u[n-2] gives g3*g0 = b0, g3*g1 = b1, g3*g2 = b2, g3*g4 = -a1, g3*g5 = -a2. Choosing g3 = 1 yields the gains in option D.*
q40	Identify minimum number of delay elements needed for a second-order IIR filter.	**Two is the minimum number of delay elements needed to realize a general second order IIR filter.**	*A second-order IIR filter has two poles, requiring at least two memory elements (delays). Direct-form II canonical realization uses exactly two delays, while the shown diagram uses four (separate input and output delays). Hence the minimum is two.*
q41	Identify filter with magnitude response near 0 at DC, rising to ~1 by Ω=π/4 and remaining ≈1 up to π.	**G_2**	*The described magnitude response is low at low frequencies and high at higher frequencies, characteristic of a high-pass filter. Among the four causal filters, G_2 exhibits this high-pass shape.*
q42	Determine the magnitude gain of filter G_3 at Ω=π/2.	**|H_3(Omega = pi/2)| = 0.98**	*From the provided magnitude plots, G_3 shows a high-pass response with unity gain in the passband. At Ω=π/2, which lies in the passband, the gain is slightly below unity due to filter roll■off, giving approximately 0.98.*
q43	Which filters could have an FIR impulse response shorter than 4 samples?	**G_1, G_2**	*For a causal FIR filter of length L driven by a unit■step input, the output reaches its steady■state value after L■1 samples. In the figure, G■ attains steady state immediately (L=1) and G■ after one sample (L=2). G■ and G■ either require more than two samples or exhibit non■linear behavior, so only G■ and G■ could be FIR filters with length <4.*
q44	Which filter could be a median filter?	**G_3**	*A causal median filter of length 3 removes isolated spikes while preserving step levels and introduces a one■sample delay. The output y■[n] shows exactly this behavior: single■sample peaks in u[n] are absent, and constant or ramp segments appear delayed by one sample, indicating G■ could be a median filter.*
q45	Phase of discrete-time transfer function at z = e^{jπ} after bilinear transform	**angle H(z) = -pi/2**	*The bilinear transform maps z = -1 to s → j∞. For H(s) = (β s + 1)/(α s^2 + s + 1), the high-frequency phase is -90° because the relative degree (denominator - numerator) is 1. Hence angle H(z) = -π/2.*
q46	Select all systems whose phase response is zero	**G_4**	*G1: FIR [1, 1/2, 1/3] gives complex H(e^{jω}). G2: first-order autoregressive yields non-zero phase. G3: pure delay gives phase -ω. G4: h[n] = δ[n+1] + 2δ[n] + δ[n-1] → H(e^{jω}) = 2 + 2cos ω, real and non■negative for all ω, thus zero phase.*
q47	Which assert ensures enough frequencies for least squares estimation of three parameters?	**assert(2*L>2);**	*Three unknowns (a1,b0,b1) require at least three equations. Each frequency provides two (real and imag) equations, so need 2L ≥ 3 → L ≥ 2. The condition assert(2*L>2) is equivalent to L≥2 for integer L.*
q48	Which expression correctly fills F(2*l+1,:) in the least squares setup?	**F(2*l+1,:) = [-R*cos(Phi-Om), 1, cos(Om)];**	*From H(e^{jΩ}) = (b0+b1 e^{-jΩ})/(1+a1 e^{-jΩ}) we derive real equation: R cosΦ + R a1 cos(Φ-Ω) - b0 - b1 cosΩ = 0. Rearranged gives [-R cos(Φ-Ω), 1, cosΩ]·[a1;b0;b1] = R cosΦ, matching G(2l+1)=R cosΦ.*

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q49	Find the DFT bin index for frequency $\Omega = \pi/4$ with N = 1000 samples.	**H_hat(pi/4) = Y_m[125] / U_e[125]**	*The DFT bin k corresponding to Ω is $k = \Omega * N / (2\pi) = (\pi/4)*1000/(2\pi) = 125$. Hence the frequency response estimate is the ratio of the DFT coefficients at bin 125.*
q50	Select the transfer function whose magnitude matches the measured values $\|H(0)\|=0.05$, $\|H(\pi/2)\|=0.39$, $\|H(\pi)\|=1.96$.	**H(z) = 1/3 (z - 1) / (z + 2/3)**	*Evaluating each candidate on the unit circle: Option D gives $\|H(0)\|\approx 0$, $\|H(\pi/2)\|\approx 0.392$, $\|H(\pi)\|\approx 2$, which is closest to the measured magnitudes.*